

MUJAHIR HUSSAIN ABBASI

+1(626)-693-2696 | mujahirabbasi@gmail.com

Los Angeles, California

[GitHub](#) | [LinkedIn](#)

SUMMARY

Experienced Data Scientist, Software & Machine Learning Engineer with expertise in developing advanced models and data analytics solutions across agriculture, technology, and consumer goods. Skilled in deep learning, computer vision, and time series forecasting, with hands-on experience in cloud platforms (AWS, Azure) and tools like Python, Pyspark, TensorFlow, and PyTorch. Proven track record of optimizing model performance, driving innovation, and delivering impactful results in projects at Brainchip, Accenture, and Tata Consultancy Services. Recognized for academic excellence and professional achievements, with a strong commitment to advancing the field of machine learning.

PROFESSIONAL EXPERIENCE

KATSH Digital ID

Jan 2025 – Present

(Katsh Digital ID is a cutting-edge platform designed to provide secure, user-friendly digital identity solutions, empowering individuals to manage and protect their personal information online)

Principal AI Full-Stack Engineer (Contract)

- Developing a Palm Authentication System with 85% accuracy using deep learning(CNN and Transformer based models) and traditional image processing algorithms.
- Built a gesture recognition model with 99% accuracy for data collection, using MediaPipe & Apple Vision.
- Built world model-based anti-spoofing to detect and block replay or fake biometric inputs.
- Developed the backend using FastAPI and MySQL, integrating machine learning capabilities.
- Build a native iOS frontend in Swift.
- Deployed machine learning models on edge devices by converting Keras models to Core ML and integrating them into Swift applications.
- Proficient in leveraging tools like Cursor, GitHub Copilot, and ChatGPT to enhance daily development workflows and productivity.
- Deployed AI models and data pipelines on AWS using S3 and EFS for storage, RDS for relational data, and ECS instances and EC2 with GPUs for model inference.
- Deployed AI models and data pipelines on Google Cloud Platform (GCP) using Cloud Storage and Filestore for storage, Cloud SQL for relational data, and Google Kubernetes Engine (GKE) for scalable model inference.
- Experienced in creating demos and engaging with clients to showcase solutions.
- Represented the team at CES 2026 by delivering a live demonstration of the product to investors, partners, and industry experts.

University of Nebraska-Lincoln

Mar 2024 – Jan 2025

AI Researcher & Software Developer (Contract)

- Conducted stress detection and classified fruits and leaves of plants using hyperspectral data.
- Performed clustering of hyperspectral images and developed an advanced data visualization web application using Streamlit.
- Built a knowledge-driven Q&A chatbot using Retrieval-Augmented Generation (RAG) with LLMs & LangChain for seamless retrieval and generation from internal knowledge bases.

- Extended functionality by wrapping RAG as a LangChain Tool, integrating SQLDatabase, API, and Python REPL tools, and deploying an Agent (AgentExecutor/LangGraph) to dynamically decide between retrieval, API calls, and computations.

Brainchip

June 2023 – Dec 2023

(A leading company in the field of artificial intelligence, specializing in neuromorphic computing to enhance machine learning and edge processing capabilities).

Machine Learning Solutions Engineer (Internship)

- Optimized Yolo v5 for compatibility with BrainChip Neuromorphic Hardware.
- Created sparse versions of the Yolo v5 model, improving latency by 10% and reducing power consumption by 20%.
- Developed a benchmarking tool to assess model performance for Yolo v5, Centernet and Vision Transformers.
- Designed and demonstrated an anomaly detection demo using a custom CNN architecture on BrainChip Hardware at CES 2024.
- Provided technical support and troubleshooting assistance for various projects.
- Delivered weekly PowerPoint presentations to keep stakeholders updated on project progress.

Accenture

Feb 2021 – Aug 2022

Data Scientist | Data Engineer | Data Analyst

- Performed data wrangling and manipulation using Python and PySpark.
- Conducted feature engineering and selection utilizing statistical methods.
- Developed proof-of-concept machine learning models on Azure cloud platforms.
- Migrated the codebase from Pandas to PySpark, reducing processing time by 90%.
- Developed a demand planning model for a Consumer-Packaged Goods (CPG) company, improving predictions by 3%.
- Built ETL data pipelines using Python, SQL, and Kusto Query Language (KQL) on Microsoft Azure.
- Created real-time and historical production dashboards for Gum Sheets using PowerBI.
- Developed a proof-of-concept for Gum defect detection using Deep Learning techniques.

Mobilewalla Information Solution Limited

Sep 2020 - Jan 2021

Data Scientist

- Developed recommendation systems leveraging Transformers and Graph Convolutional Networks.
- Built a proof-of-concept (POC) for mobile device classification (Prepaid & Postpaid) using machine learning techniques.

Tata Consultancy Services

Mar 2019 - Aug 2020

Business Analyst | Data Scientist | Data Engineer

- Collected and processed data from various sources, performing joins and preparing data for model input using Pandas and PySpark.
- Led a team of two, coordinating tasks and resolving any challenges that emerged.
- Generated and presented comprehensive reports to stakeholders using Microsoft Excel.
- Developed a machine learning-based demand prediction solution on Azure Databricks, improving accuracy by 5% through feature engineering.
- Built an ETL pipeline with PySpark and developed NLP-based logic for TATA SUPERAPP.
- Analyzed client business needs and delivered insights through ad-hoc analysis.

AiSee Technologies

Mar 2018 - Feb 2019

Machine Learning Engineer & Software Engineer

- Developed a data annotation tool using Python, Tkinter, and Matplotlib, leading the data annotation process.
- Built an advanced ATM Security System and Face Recognition System by pretraining and fine-tuning YOLO models.
- Pretrained and fine-tuned the YOLOv2 model on a custom dataset using local GPUs to automatically count and classify traffic.

TECHNICAL SKILLS

Languages: Python, R, Shell, Swift, SQL, Spark, Kusto Query Language.

Libraries: Scikit-Learn, OpenCV, Numpy, Pandas, Matplotlib, Scipy, TensorFlow, Keras, PyTorch, Langchain, Spark MLLib, DAX, CUDA, RESTAPI.

Cloud: AWS stack: S3, EMR, EC2, ECS MS Azure Stack: DataBricks, Datalake Store, Blob, Data Explorer, Azure Functions, Data Factory.

Databases: Mysql, Azure SQL database, SQL Alchemy.

Data Tasks : Extraction, Cleansing, Processing, Feature Engineering, Ingestion, Reporting, Visualization, Evaluation Techniques.

Algorithms: Lasso & Ridge Regression, Logistic Regression, PCA, Decision Tree, XGBoost, Random Forest, SVM, Cluster Analysis, CNN, RNN, GNN, GAN, Transformers, ANOVA, Chi-Squared, T-test, Z-test.

Architectures: Mask RCNN, YOLOv2, Tiny YOLOv3, YOLOv4, MobileNet SSD, YOLOv5.

Software Development Tools: Agile Methodologies, Scrum, Jira, CI/CD, Deployment, GIS software, Gitlab, Microsoft Excel, StreamLit, PowerBI, GitHub, Visual Studio Code, Microsoft PowerPoint, SharePoint, Data Structures & Algorithms, SSH, Docker, Jupyter Notebooks, Apache Zeppelin.

Domain: Business Intelligence, Object Oriented Programming, Recommendation System, Time Series Forecasting, Optimization, Transfer Learning, Quantization, Statistics, A/B Testing, Edge, LLMs, Computing, Deep Learning, MLOps, Generative AI, Agentic AI.

Operating System: Linux, Unix, Windows.

Industry: Retail, CPG, Ecommerce, Education, Transportation, Government Organization, Semiconductor Manufacturing, Supply Chain & Operations, Telecommunications and Advertising.

EDUCATION

California State University, Los Angeles

Master of Science in Computer Science – Grade 3.95/4.0 | Los Angeles, California

UPWORK PROJECTS :

Agentic AI System for Sponsor Analytics

- Built an agentic AI system to detect sponsor logos in broadcast videos.
- Computed sponsor appearance timing, on-screen duration, spatial position, coverage, and visibility quality from broadcast videos.

MaizeLLM

- Developed MaizeLLM for maize disease detection and plant health assessment using LoRA fine tuning.
- Built maize datasets and converted image labels into text data for healthy and diseased plant classification.

Databricks Genie Migration Automation

- Built an MLOps automation workflow to support Databricks Genie migration across multiple workspaces.
- Used Ansible playbooks to automate configuration setup, workspace preparation, and deployment tasks.
- Reduced manual migration effort and improved consistency across development, staging, and production environments.

ACHIEVEMENTS/CO-CURRICULAR

- Received Appreciation Email from Principal Director at Accenture & “On The Spot Award” and “Appreciation Note” in TCS.
- Awarded the Non-Resident Tuition Fee Waiver for outstanding academic performance by California State University, Los Angeles.
- “Detection of Surface Water from Satellite Imagery using Deep Learning with Indirect Proxy Based Label Collection Method.” [IEEE](#)
- Program Committee Member ICAIML AE 2026 and Peer Reviewer for IEEE, Springer Nature and BIMA.